

Generation Interconnection System Impact Study Report

For Queue Project Q15-0527

Vergennes 9 345 kV

October 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0527. The Interconnection Customer proposes a 1110.7 MW natural gas combined cycle facility interconnecting at the Vergennes 9 345 kV substation in the West Virginia area. The request was received on October 20, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 190 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 16 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$76,765,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VERGENNES 9 345/138 kV XFMR	HAWKS NEST HYDRO – VERGENNES 9 345 KV	161.6	469	69.6
KEWEENAW SUBSTATION – VERGENNES 9 138 KV	HAWKS NEST HYDRO – VERGENNES 9 345 KV	148.7	224	35.3
STONE CORRAL – KEWEENAW 138 KV	HAWKS NEST HYDRO – VERGENNES 9 345 KV	131.9	246	35.3
GLEN LAKE – DORRIS 138 KV	COULEE AVE – AMOS 345 KV	128.3	178	5.4
BIG FISH – TEAL 138 KV Line	COULEE AVE – AMOS 345 KV	111.4	222	5.8
BERY MTN – RIO GRANDE 138 KV	SCARBRO – WASTEWATER SUBSTATION 138 KV line	107.0	204	5.8
BEECH RIDGE ENERGY LLC 138/1kV Transformer Bank	HAWKS NEST HYDRO – VERGENNES 9 345 KV L	106.3	347	30.4
VERGENNES 9 – EAST HOLBROOK 138 KV Line	SCARBRO SUBSTATION – WASTEWATER 138 KV	105.2	229	5.8
BEURY MTN – PRODUCTOS ELECTRICOS 2 138 KV	WASTEWATER – SCARBRO 138 KV L	104.9	220	5.8
OSCEOLA SOLAR FACILITY – OYSTER BAY 138 KV	SCARBRO – WASTEWATER 138 KV	104.7	226	5.8
EAST HOLBROOK – OSCEOLA SOLAR FACILITY 138 KV	WASTEWATER – SCARBRO 138 KV Line	104.5	228	5.8
MATTOON EAST – ROCKAWAY BEACH 138 KV Line	COULEE AVE – SPORN 345 KV	104.4	255	6.2
PRODUCTOS ELECTRICOS 2 – OYSTER BAY 138 KV L	SCARBRO – WASTEWATER 138 KV Line	104.1	223	5.8

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ROCKAWAY BEACH – HAMPSHIRE PAPER 138 KV	COULEE AVE – SPORN 345 KV	103.4	255	6.2
DORRIS – MALAD 138 KV	AMOS – COULEE AVE 345 KV	101.3	231	6.5
ALPHA RD – PAWNEE 138 KV Line	BEECH RIDGE ENERGY LLC 138/1 kV Transformer	100.1	154	8.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VERGENNES 9 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$11,910,000	100% (\$11,910,000)
KEWEENAW SUBSTATION – VERGENNES 9 138 KV	Rebuild line with higher-capacity conductor	\$2,106,000	100% (\$2,106,000)
STONE CORRAL – KEWEENAW SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$22,253,000	100% (\$22,253,000)
GLEN LAKE – DORRIS 138 KV	Rebuild line with higher-capacity conductor	\$21,468,000	30% (\$6,389,000)
TEAL – BIG FISH 138 KV	Rebuild line with higher-capacity conductor	\$15,157,000	30% (\$4,562,000)
RIO GRANDE – BEURY MTN 138 KV L	Reconductor line	\$6,102,000	100% (\$6,102,000)
BEECH RIDGE ENERGY LLC 138/1kV Transformer Bank	Replace transformer with higher-rated unit	\$7,682,000	100% (\$7,682,000)
VERGENNES 9 – EAST HOLBROOK 138 KV Line	Reconductor line	\$3,038,000	100% (\$3,038,000)
PRODUCTOS ELECTRICOS 2 – BEURY MTN 138 KV	Reconductor line	\$3,116,000	100% (\$3,116,000)
OSCEOLA SOLAR FACILITY – OYSTER BAY 138 KV L	Reconductor line	\$2,441,000	100% (\$2,441,000)
OSCEOLA SOLAR FACILITY – E HOLBROOK 138 KV	Reconductor line	\$1,346,000	100% (\$1,346,000)
MATTOON EAST – ROCKAWAY BEACH 138 KV Line	Reconductor line	\$2,633,000	36% (\$942,000)
PRODUCTOS ELECTRICOS 2 – OYSTER BAY 138 KV	Reconductor line	\$1,998,000	100% (\$1,998,000)
ROCKAWAY BEACH – HAMPSHIRE PAPER 138 KV	Reconductor line	\$4,349,000	36% (\$1,555,000)
DORRIS – MALAD 138 KV	Reconductor line	\$1,242,000	26% (\$321,000)
PAWNEE – ALPHA RD 138 KV	Reconductor line	\$1,004,000	100% (\$1,004,000)
	Total allocated to project		\$76,765,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.